Tutorial-7

A moving streaming service wants to classify movies into Categories : Action or romance using a kNN classifier

Input Features

Action scenes count - No. of action sequences
Romance scenes count - No of romantic sequences

Target Variables

Genere (Action = A , Romance = R)

Training Dataset

~~Movie ID~~

Movie ID	Action scene	Romance scene	Genere
1	15	2	A
2	10	12	R
3	18	1	A
4	2	14	R
5	14	3	A
6	3	11	R
7	16	2	R
8	2	13	R

Test Data (with actual labels)

Test ID	Action scene	Romance scene	Actual genre
T_1	10	5	A
T_2	4	10	R
T_3	17	2	A

1. Calculate the Eudidean distance b/w test instanc T_1 (Action=10, Romance=5) and all training samples show your complete calculations.

2. classify all 6 test instances using KNN with k=1, 3, 5 create a prediction table showing predicted genre for each test instance at different k values.

3. calculate the classification accuracy for each k value of k=1,3,5 formula: Accuracy = $\frac{\text{No. of correct predictions}}{\text{Total test predictions}} \times 100\%$

4. Create confusion matrix for k=3 showing

5. calculate precision - recall and F1 - score for the action call when k=3

Euclideain formula:- $D = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$

$(15,2) = \sqrt{(10-15)^2+(5-2)} = \sqrt{25+9} = \sqrt{34} = 5.83$

$(1,12) = \sqrt{(10-1)^2+(5-12)^2} = \sqrt{81+49} = \sqrt{130} = 11.4$

$(18,1) = \sqrt{(10-18)^2+(5-1)^2} = \sqrt{64+16} = \sqrt{80} = 8.94$

$(2,14) = \sqrt{(10-2)^2+(5+14)^2} = \sqrt{64+81} = \sqrt{145} = 12.04$

$(14,3) = \sqrt{(10-14)^2+(5-3)^2} = \sqrt{16+4} = \sqrt{20} = 4.47$

$(3,11) = \sqrt{(10-3)^2+(5-11)^2} = \sqrt{49+36} = \sqrt{85} = 9.21$

$(16,2) = \sqrt{(10-16)^2+(5-2)^2} = \sqrt{36+9} = \sqrt{45} = 6.70$

$(2,13) = \sqrt{(10-2)^2+(5+13)^2} = \sqrt{64+64} = \sqrt{128} = 11.31$

movieID	Distance	Genre
5	4.817	A
1	5.38	A
7	6.70	A

Classification of T_1

when k=1 Nearest Neighbours (5, 1, 7) Genre A prediction (A)

(x, y) (4, 10)

$(15,2) = \sqrt{(4-15)^2+(10-2)^2} = \sqrt{121+64} = \sqrt{185} = 13.6$

$(1,12) = \sqrt{(4-1)^2+(10-12)^2} = \sqrt{9+4} = \sqrt{13} = 3.66$

$(18,1) = \sqrt{(4-18)^2+(10-1)^2} = \sqrt{196+81} = \sqrt{277} = 16.6$

$(2,14) = \sqrt{(4-2)^2+(10-14)^2} = \sqrt{4+16} = \sqrt{20} = 4.47$

$(14,3) = \sqrt{(4-14)^2+(10-3)^2} = \sqrt{100+49} = \sqrt{149} = 12.20$

$(3,11) = \sqrt{(4-3)^2+(10-11)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$

$(16,2) = \sqrt{(4-16)^2+(10-2)^2} = \sqrt{144+64} = \sqrt{208} = 14.4$

$(2,13) = \sqrt{(4-2)^2+(10-13)^2} = \sqrt{4+9} = \sqrt{13} = 3.60$

Movie Id	Distance	Genre
6	1.41	R
2	3.60	R
8	3.60	R

$(17,2)$

$(15,2) = \sqrt{(17-15)^2+(2-2)^2} = \sqrt{4+0} = 2$

$(1,12) = \sqrt{(17-1)^2+(2-12)^2} = \sqrt{256+100} = 18.8$

$(18,1) = \sqrt{(17-18)^2+(2-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4$

$(2,14) = \sqrt{(17-2)^2+(2-14)^2} = \sqrt{225+144} = 19.2$

$(14,3) = \sqrt{(17-14)^2+(2-3)^2} = \sqrt{9+1} = \sqrt{10} = 3.16$

$(3,11) = \sqrt{(17-3)^2+(2-11)^2} = \sqrt{196+81} = \sqrt{277} = 16.6$

$(16,2) = \sqrt{(17-16)^2+(2-2)^2} = \sqrt{1+0} = \sqrt{1} = 1$

$(2,13) = \sqrt{(17-2)^2+(2-13)^2} = \sqrt{225+121} = \sqrt{346} = 18.60$

Movie ID	Distance	Genre
7	1	A
3	1.4	A
1	2	A

Actual labels vs predicted labels

Test ID	Actual	predicted (k=1)	predicted (k=2)
T_1	A	A	A
T_2	R	R	R
T_3	A	A	A

All predictions match with actual labels

3.- Accuracy = $\frac{\text{Correct predictions}}{\text{Total predictions}}$

$= \frac{3}{3} \times 100 = 100\%$ Accuracy.

4. confusion matrix

Action → positive

Romance → Negative

	A	R
A	2	0
R	0	1

TP = 2

FP = 0

FN = 0

TN = 1

5. precision = $\frac{TP}{(TP+FP)} = \frac{2}{2+0} = 1$

Precision = 100%.

Recall = $\frac{TP}{(TP+FN)} = \frac{2}{2+0} = 1$

Recall = 100%.

F1 Score

$F_1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$

$F_1 = \frac{2 \times (1 \times 1)}{1+1} = 1$

F1 Score